PHYSICAL CLIMATE RISK: FLOOD MODEL COMPARISON, HAZARD AND VULNERABILTY ASSESSMENT AND FINANCIAL IMPLICATIONS

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Master Thesis

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Abstract

Flooding is one of the most frequent and damaging natural hazards globally, and its risk is projected to increase due to climate change. Accurate flood risk assessments are essential for spatial planning, climate adaptation, and ESG-aligned financial decision-making. This thesis investigates physical flood risk by comparing global and local hazard models, evaluating standardized vulnerability functions, and quantifying financial implications for real estate assets in Slovenia.

Two flood hazard models are assessed: the global WRI Aqueduct River Flood model and the high-resolution local model Integralna karta globin (IKG). A geospatial validation dataset comprising flood depth and damage records from four flood events in Slovenia (2010–2023) is used to evaluate model accuracy. Flood depths are analyzed using both regression and classification metrics, while vulnerability is assessed through comparison of observed damages against standardized depth–damage functions provided by the European Commission Joint Research Centre (JRC).

A key technical contribution is the development of a data processing pipeline that harmonizes heterogeneous IKG hazard data and integrates it into the modular Physrisk platform using the Zarr data format. This enables scalable, scenario-based risk assessment and model intercomparison.

Findings suggest that while local models offer improved accuracy for continuous flood depth prediction in regression tasks, the global model performs better in categorical classification. Proximity to water bodies was not found to significantly improve prediction accuracy. Depth–damage functions align well with observed damage distributions for residential properties, though segmentation by property type reveals variability in model performance.

The thesis contributes to the validation and integration of physical climate risk models for real estate applications and enables more accurate, location-specific flood risk estimates to support planning and investment decisions.

Povzetek

Povzetek v slovenščini naj ne bo daljši od ene strani.

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List of Algorithms

Abbreviations

JSI ... Jožef Stefan Institute
IPS ... International Postgraduate School
AEP ... Average Exceedance Probability
EAD ... Expected Annual Damage
ESG ... Environmental, Social, and Governance
JRC ... Joint Research Centre
RP ... Return Period
WRI ... World Resources Institute

Symbols

$j^\star$ ... black-body irradiance
σ ... Stefan's (or Stefan-Boltzmann) constant

Glossary

Glossary of terms

Chapter 1

Introduction

1.1 Background and Motivation

Flooding is one of the most frequent and damaging natural hazards globally, responsible for extensive economic losses, social disruption, and loss of life. Climate change is expected to further intensify these risks by increasing both the frequency and severity of hydrological extremes. The Intergovernmental Panel on Climate Change (IPCC) projects a growing incidence of intense precipitation events and flood-related disasters, particularly in regions affected by higher hydrological variability and sea-level rise [1].

In Europe, recent catastrophic flood events—such as those in Germany and Belgium in 2021, and in Slovenia in 2023—have exposed significant vulnerabilities in infrastructure, spatial planning, and emergency preparedness. According to the European Environment Agency (EEA), riverine and pluvial floods already account for more than one-third of total weather-related losses in Europe, a share that is projected to increase over coming decades without improved mitigation and adaptation measures [2].

These developments underscore the urgent need for robust and spatially detailed flood risk models that can support risk-sensitive decision-making in domains such as insurance pricing, urban planning, asset valuation, and regulatory compliance. This is particularly critical for stakeholders in the real estate and financial sectors, who are increasingly required to assess and disclose exposure to climate-related physical risks as part of their environmental, social, and governance (ESG) reporting obligations [3], [4].

1.2 Climate Risk and ESG Disclosure

Climate risk refers to the potential adverse effects of climate change on economic systems, institutions, and physical assets. It is commonly classified into three interrelated categories: *physical*, *transition*, and *liability* risks [5].

- **Physical risks** arise from the direct consequences of climate variability and long-term climate change. These include acute hazards such as floods, storms, and wildfires, as well as chronic effects such as sea-level rise and shifting temperature trends. These risks can cause immediate physical damage and long-term degradation of asset value.

- **Transition risks** arise from the transition to a low-carbon economy. They include policy changes (e.g., carbon pricing, emissions regulations), technology disruption, evolving market preferences, and reputational risks associated with inaction or misalignment.

- **Liability risks** are considered by some [6] and encapsulate risks arising when those affected by anthoropogenic climate change seek compensation.

These categories are increasingly embedded in ESG-aligned regulatory frameworks such as those developed by the Task Force on Climate-related Financial Disclosures (TCFD) and the Network for Greening the Financial System (NGFS). In the European context, the European Central Bank (ECB) requires significant institutions to assess, manage, and disclose climate-related and environmental risks, including physical risks. According to the ECB's supervisory guidance, banks are expected to integrate these risks into their internal governance, risk management, stress testing, and disclosure practices [3].

Physical risk is typically modeled as the interaction of three fundamental components:

- **Hazard** – the probability and intensity of a climate-related event (e.g., flood depth for a 100-year return period),
- **Vulnerability** - the expected damage to an asset given the intensity of the hazard,
- **Exposure** - the presence and value of assets located in potentially affected areas.

These components form the foundation of physical climate risk models used in both regulatory compliance and voluntary disclosures. However, despite the wide adoption of this conceptual framework, practical implementation often suffers from limited model transparency, inconsistent spatial resolution, and poor validation—especially at the local scale relevant to property-level investment and risk management.

This thesis focuses specifically on **acute physical risks**, with a particular emphasis on riverine flooding and its financial impact on real estate assets. By comparing the performance of global and local flood models and evaluating standardized vulnerability functions against observed data, the work aims to contribute to more accurate and transparent risk quantification in support of ESG-aligned decision-making.

1.3 Problem Statement

While global flood hazard models such as the WRI Aqueduct River Floods provide broad coverage and climate scenario integration, they are often limited in spatial resolution and may not reflect local hydrological conditions. On the other hand, high-resolution local models, such as Slovenia's *Integralna karta globin* (IKG), offer detailed inundation mapping but are typically fragmented, static (i.e., not climate scenario-aware), and difficult to integrate into broader risk platforms.

Furthermore, flood vulnerability is typically modeled using generalized depth–damage functions derived from empirical or synthetic data. These functions may perform well in aggregate, but their applicability across different building types, geographies, and damage-reporting standards remains uncertain. This creates uncertainty in downstream financial estimates and limits the usefulness of such models for asset-level ESG disclosure.

1.4 Research Objectives

The purpose of this thesis is to evaluate how well existing flood models can represent real-world flood depths and damage outcomes in Slovenia. Specifically, it compares a global hazard model (WRI Aqueduct) and a high-resolution local model (IKG) to assess the accuracy of their flood depth predictions.

Separately, the thesis tests global depth-damage functions, used to estimate flood-related property value loss, against reported damage data, with the goal of evaluating and potentially improving their performance. Together, these analyses aim to assess the reliability of current modeling tools for understanding physical climate risk and supporting property related decision making in the context of increasing flood exposure.

The main goals of the research are:

1. To improve existing physrisk risk assessment framework with the following contributions:
   (a) Develop a data processing pipeline for creating a unified and homogenous SI IKG flood depth model for different areas of importance.
   (b) Develop a data onboarding pipeline for integrating this new unified model into physrisk framework.
2. To compare the performance of a global flood hazard model (WRI Aqueduct) and a local Slovenian model (IKG) using real flood data.
3. To evaluate how model resolution and proximity to water bodies affect prediction accuracy.
4. To assess the accuracy of global flood depth–damage functions on reported damage cases in Slovenia and improve their performance.

1.5 Structure of the Thesis

The remainder of the thesis is structured as follows:

- **Chapter 2** presents a review of relevant literature and related work on flood hazard modeling, vulnerability estimation, and ESG-aligned risk frameworks.
- **Chapter 3** describes the study area, datasets, and the methodological framework used for model evaluation and data processing.
- **Chapter 4** details the integration of the local IKG model into the Physrisk framework and presents the architecture of the comparison toolset.
- **Chapter 5** provides experimental results, including quantitative evaluation of model performance, depth–damage function analysis, and financial impact estimates.
- **Chapter 6** discusses the key findings, limitations, and possible future research directions, and concludes the thesis.

Chapter 2

Literature Review and Related Work

This chapter reviews key research and frameworks related to flood risk modeling, with a focus on hazard modeling, vulnerability assessment, exposure estimation, and regulatory context. The literature spans hydrology, geospatial modeling, and financial risk assessment, with special emphasis on tools applicable to European and Slovenian contexts.

2.1 Flood Hazard Modeling

Flood hazard modeling involves estimating the spatial extent and intensity of flood events for given probabilities or return periods. It forms the foundational input to any flood risk assessment, determining where and how deep flooding occurs under specific scenarios. Two main categories of models are commonly used: global models and high-resolution local hydraulic models.

2.1.1 Global Flood Models

Global flood models are designed for scalability and global coverage. They typically simulate flood hazard using simplified hydrological and hydraulic models driven by global datasets such as digital elevation models, river discharge records, and climate projections. A widely used example is the **WRI Aqueduct River Flood model**, which provides fluvial flood depth maps for return periods ranging from 2 to 1000 years, with a spatial resolution of 1 km and support for climate change scenarios such as RCP4.5 and RCP8.5 [7], [8].

These models are designed to support large-scale comparisons, stress testing, and financial disclosures. However, their coarse spatial resolution limits their application in local-level asset risk estimation or infrastructure planning [1], [6].

2.1.2 Local Hydraulic Models

In contrast, local hydraulic models such as Slovenia's **Integralna karta globin (IKG)** are developed with detailed topography, land use, and hydraulic calibration. These models often rely on 1D or 2D hydraulic simulations and historical flood data to generate detailed flood hazard maps for selected municipalities. In the case of IKG, flood depth is categorized into discrete classes (GM, GS, GB), corresponding to increasing flood severity, and available for return periods of 10, 100, and 500 years [9], [10].

While local models lack global coverage and scenario flexibility, they provide high spatial precision (typically 5–10 m resolution) that is critical for building-level flood risk modeling and regulatory planning.

2.1.3 Return Periods and Exceedance Probabilities

Both global and local models represent flood hazard as a function of return period T, which reflects the average time between flood events of a given magnitude. The annual exceedance probability (AEP) is defined as:

$$p = \frac{1}{T}$$

This mathematical formulation allows different hazard layers to be harmonized and aggregated in subsequent risk calculations [11].

Recent studies show that the choice and granularity of return periods significantly affect total risk estimates—especially for metrics like Expected Annual Damage (EAD)—emphasizing the importance of high return-period resolution [11].

2.1.4 Climate Scenarios and Hazard Projections

Flood models increasingly integrate climate scenarios to account for the effects of climate change on future flood hazard. The Representative Concentration Pathways (RCPs) developed by the IPCC serve as standardized radiative forcing trajectories:

- **RCP4.5** assumes stabilization of greenhouse gas emissions, leading to approximately 2.4–3.2°C warming by 2100.
- **RCP8.5** reflects continued high emissions, potentially leading to >4°C warming [1].

Models like WRI Aqueduct simulate future flood depths under both RCP scenarios, allowing risk analysts to quantify how increasing precipitation, runoff, and sea level may affect hazard exposure in coming decades [1], [12].

2.1.5 Summary

Global models offer scalability and scenario coverage but often lack spatial detail. Local models, like Slovenia's IKG, offer high resolution and contextual accuracy but lack global comparability and scenario flexibility. The choice of model depends on the application—portfolio stress testing may prefer global models, while asset-level analysis or planning benefits from detailed local models [1], [13].

2.2 Vulnerability Modeling

Flood vulnerability quantifies the relationship between hazard intensity (typically flood depth) and the resulting damage to physical assets. It is commonly modeled using **depth–damage functions** (DDFs), which map flood depth to fractional loss of asset value.

2.2.1 Types of Depth–Damage Functions

Depth–damage functions can be derived through various approaches:

- **Empirical DDFs** are constructed from post-disaster damage surveys and insurance claims. While highly context-specific, their availability is often limited due to data confidentiality and geographic constraints.
- **Synthetic DDFs** are developed via expert judgment, simulations, or engineering heuristics. These are common in regions without historical damage data and are often used in regulatory frameworks (e.g., HAZUS in the U.S.) [14].

- **Standardized global DDFs** offer harmonized vulnerability curves across countries and building types. A notable example is the dataset by the Joint Research Centre (JRC), which includes normalized depth–damage functions and maximum damage values for various assets globally [13].

These functions typically start at a threshold depth (e.g., 0.1 m), increase with rising floodwater, and saturate at a maximum loss value (e.g., 70–90% of replacement value for residential buildings). An example is shown in Figure 3.2 in Chapter 3.

2.2.2 Use of JRC Vulnerability Curves

This thesis uses JRC standardized vulnerability functions due to their consistency and availability for European countries. The curves are calibrated to represent fractional damage relative to building replacement value (BRV), making them suitable for property-level assessments across regions [13].

JRC functions are stratified by asset type (residential, commercial, industrial) and were assigned to Slovenian buildings based on available metadata or assumed building usage types in the damage reports.

2.2.3 Validation and Regional Applicability

While standardized functions facilitate large-scale modeling, their accuracy at local levels is not guaranteed. Validation with empirical data is essential.

In this thesis, local flood damage reports from Slovenia (2010–2023) were used to test the applicability of JRC curves [15]. Reported flood depths and damage amounts were compared to DDF-predicted losses to assess:

- The bias and variance of model predictions,
- Whether adjustments are needed for specific building types or flood classes,
- How well the JRC curves transfer to Central European urban and rural contexts.

This approach aligns with recent recommendations on validating vulnerability models using locally observed events and supports adaptation of standardized curves to regional contexts [16].

2.2.4 Uncertainty and Transferability

One of the key limitations in flood vulnerability modeling is the uncertainty in applying DDFs across different geographies and asset types. Factors such as building materials, maintenance standards, and floor elevation significantly influence damage outcomes but are rarely captured in standardized functions [17].

This thesis adopts a conservative approach by applying DDFs to estimate relative (not absolute) damage and focuses primarily on model comparison and validation, rather than financial forecasting. This helps reduce the impact of scale errors and exposure inaccuracies.

2.3 Exposure Modeling and Financial Risk Metrics

Flood exposure refers to the value and spatial distribution of assets that are potentially affected by flooding. In the context of physical climate risk, it typically denotes buildings, infrastructure, or other economic assets located in flood-prone areas.

2.3.1 Asset-Level Exposure Estimation

Accurate exposure estimation is essential for risk quantification. Methods vary based on data availability and spatial resolution:

- **Cadastral or administrative data** include tax records, property registries, or insurance databases that provide building location and value. These offer the most accurate source of asset-level exposure, but are often limited due to privacy or accessibility constraints.
- **Proxy-based estimates** use geospatial data (e.g., building footprints, land use maps) and assign average values per m^2 based on construction type or region. These are useful for large-scale or cross-country modeling where granular data is unavailable.
- **Synthetic approaches** simulate building distribution and value using statistical models calibrated on available reference areas. These are common in flood modeling platforms such as HAZUS or GLOFRIS [14], [18].

For this thesis, property values are estimated based on building location and typology, harmonized with the vulnerability curves from JRC [13]. Exposure is expressed in relative terms (percentage damage), reducing dependence on exact monetary valuations.

2.3.2 Building Replacement Value (BRV) as Exposure Metric

Many flood loss models define damage relative to the **Building Replacement Value** (BRV), i.e., the cost to fully rebuild the structure. This standardization allows for comparability across countries and buildings of different ages or materials [13], [17].

BRV-based modeling simplifies vulnerability assignment and integrates well with DDFs. However, it requires assumptions about construction costs and asset characteristics, which introduces uncertainty when applied at national or continental scales.

2.3.3 Financial Risk Metrics: Expected Annual Damage (EAD)

Expected Annual Damage (EAD) is a commonly used financial risk indicator that captures long-term expected flood losses by integrating damage across all modeled return periods.

EAD is computed as the area under the exceedance probability-loss curve (risk curve):

$$\text{EAD} = \int_0^1 D(h(p)) \cdot E \, dp$$

This approach is widely recommended in regulatory and scientific literature as it accounts for both frequent low-intensity floods and rare extreme events [11], [12].

EAD is used by:

- **Banks and insurers** to quantify flood exposure in portfolios,
- **ESG reporting platforms** (e.g., OS-Climate Physrisk) to assess asset-level physical risk,
- **Regulators** such as the ECB, EBA, and NGFS as part of climate stress testing frameworks [3], [4].

2.3.4 Relevance for ESG and Climate Stress Testing

Financial institutions are increasingly required to disclose physical climate risk under TCFD, ECB, and NGFS guidelines [3], [4]. EAD and related metrics are key components of these frameworks.

The Physrisk platform used in this thesis supports standardized EAD estimation at asset-level scale and facilitates integration with broader ESG and financial systems [5]. This aligns with emerging best practices in climate risk management and scenario analysis.

2.4 Model Integration and Comparison Frameworks

Integrating hazard, vulnerability, and exposure components into a unified flood risk framework requires both conceptual alignment and technical infrastructure. In recent years, several open-source and proprietary platforms have emerged to address this challenge by providing modular systems for flood risk modeling, evaluation, and disclosure.

2.4.1 Integration of Heterogeneous Data Sources

Flood risk assessments often rely on heterogeneous data inputs, including:

- Rasterized flood hazard models with different resolutions and coordinate systems (e.g., WRI Aqueduct in EPSG:4326 vs. IKG in EPSG:3794).
- Depth–damage functions from standardized or region-specific sources (e.g., JRC [13], HAZUS [14]).
- Asset-level exposure layers with variable spatial precision and metadata completeness.

Standardizing these inputs requires preprocessing pipelines to harmonize coordinate systems, spatial resolution, and return period semantics. The use of intermediate formats such as **Zarr** (cloud-native chunked arrays) and **GeoTIFF** enables efficient storage, access, and cross-model evaluation, as demonstrated in the OS-Climate Physrisk platform [5].

2.4.2 Physrisk Platform Architecture

Physrisk is an open-source framework developed under the OS-Climate initiative. It enables modular evaluation of physical climate risks by separating the hazard, vulnerability, and financial impact components into composable modules. Hazard indicators are ingested via the `os-risk-hazard` pipeline, standardized as Zarr arrays indexed by latitude, longitude, and return period.

Once onboarded, hazard layers can be combined with vulnerability functions and exposure estimates to compute risk indicators like Expected Annual Damage (EAD). The outputs are accessible through a dedicated API and used in ESG disclosure tools and scenario analysis workflows [5].

2.4.3 Model Comparison Methodologies

As flood hazard models vary widely in resolution, methodology, and assumptions, comparison frameworks are essential to evaluate:

- **Prediction accuracy** of flood depths and damage estimates,

- **Model bias** due to resolution or proximity to water bodies,
- **Robustness** under different climate scenarios or return period resolutions.

Model comparison studies often use regression metrics (e.g., RMSE, MAE) or classification accuracy across hazard classes (e.g., <0.5 m, 0.5–1.5 m, >1.5 m). Local validation datasets—such as flood damage reports collected by civil protection agencies—enable empirical benchmarking [15], [18].

Studies such as [11] have emphasized the importance of consistent return period selection and integration ranges when comparing EAD across models. Uncertainty from different modeling assumptions can be bounded using ensemble approaches or sensitivity analysis [17].

2.4.4 Model Portability and ESG Integration

The use of open, modular architectures like Physrisk enables integration of regionally calibrated models (e.g., IKG) alongside global datasets (e.g., WRI Aqueduct [7]). This flexibility is increasingly important for:

- **Financial institutions** performing climate scenario analysis at portfolio scale,
- **Developers** building ESG platforms with physical risk indicators,
- **Regulators** requiring harmonized risk evaluation under stress testing frameworks [3], [4].

Comparison-ready architectures support repeatable workflows and transparency, both of which are emphasized in regulatory guidance and scientific literature [12].

2.5 Regulatory and Institutional Context

Flood risk is not only a scientific and technical issue but also a key component of financial regulation, urban planning, and climate adaptation policy. In recent years, regulatory bodies and international institutions have increasingly mandated the integration of physical climate risk—especially flood risk—into financial disclosures, risk management processes, and stress testing exercises.

2.5.1 Climate Risk Disclosure Frameworks

A major milestone in climate risk governance was the establishment of the **Task Force on Climate-related Financial Disclosures (TCFD)** in 2015. Its final recommendations, published in 2017, laid the foundation for disclosing physical and transition risks under four pillars: governance, strategy, risk management, and metrics & targets [4]. TCFD recommends scenario-based assessments of physical risks such as river flooding under different Representative Concentration Pathways (RCPs).

In Europe, the TCFD framework has been integrated into supervisory expectations issued by the **European Central Bank (ECB)**. Its 2020 guide outlines how banks are expected to identify, assess, and disclose climate-related and environmental risks in their portfolios, including physical hazards such as floods [3]. Institutions are encouraged to develop internal methodologies for quantifying risk exposure and potential financial loss under climate scenarios.

2.5.2 Stress Testing and Portfolio-Level Risk

The ECB, European Banking Authority (EBA), and Network for Greening the Financial System (NGFS) have launched climate stress tests to assess how banks and insurers would be affected by physical risks under both baseline and extreme climate trajectories. These stress tests require:

- Detailed spatial modeling of physical hazards (e.g., flood depth),
- Scenario-aware risk metrics (e.g., Expected Annual Damage),
- Asset-level exposure and vulnerability mapping.

The ECB's 2022 Climate Stress Test highlighted limitations in banks' ability to assess physical risks due to a lack of granular hazard data and standardized methodologies. This finding reinforces the need for transparent and reproducible frameworks such as OS-Climate's Physrisk, which allow banks and ESG platforms to onboard regional hazard datasets and harmonize them with portfolio data.

2.5.3 Guidance from Research Institutions and Think Tanks

Reports from international research groups and climate-finance think tanks have emphasized the importance of integrating asset-level physical climate risk into investment and adaptation planning. For instance, Bavandi et al. [12] outline lessons for modeling extreme weather scenarios in emerging markets, emphasizing the need for detailed spatial data and alignment with RCP-based climate pathways.

Similarly, the European Environment Agency (EEA) encourages cities and regions to adopt evidence-based approaches to climate adaptation and flood risk management, including geospatial risk mapping and local vulnerability modeling [2].

2.5.4 Role of National Authorities

At the national level, ministries of the environment and civil protection agencies—such as the **Uprava RS za zaščito in reševanje** in Slovenia—are responsible for developing flood hazard maps, collecting damage reports, and designating flood risk zones under the EU Floods Directive. These data and designations serve as critical inputs for both public and private sector climate risk assessments [9], [15].

Moreover, national legislation (e.g., the Slovenian regulation on flood zones [19]) defines how land parcels are classified into flood hazard classes, affecting insurance premiums, development rights, and adaptation investment decisions.

2.5.5 Implications for Flood Risk Modeling

The convergence of scientific, regulatory, and financial perspectives has made it essential to develop flood risk models that are:

- Technically rigorous and scenario-aware,
- Spatially granular and asset-level,
- Interoperable with disclosure and stress-testing platforms.

These requirements motivate the work presented in this thesis, which aims to harmonize local hazard data, validate vulnerability assumptions, and support risk quantification workflows consistent with regulatory expectations and institutional best practices.

2.6 Summary

This chapter reviewed the academic literature and institutional context surrounding flood risk modeling, with a focus on four core components: flood hazard simulation, vulnerability assessment, exposure estimation, and model integration within financial and regulatory frameworks.

Flood hazard modeling has progressed from detailed, site-specific hydraulic simulations to global, scenario-aware datasets such as Aqueduct and JRC river flood models, which enable spatially explicit assessments across a range of return periods and climate scenarios [8], [18], [20]. While global models offer coverage and consistency, they trade off spatial resolution and local accuracy—an issue addressed by high-resolution national models such as Slovenia's *Integralna karta globin* (IKG).

Vulnerability assessment remains centered on depth–damage functions, notably those developed by the Joint Research Centre (JRC) [13]. These functions offer global standardization, but their regional transferability and calibration to local damage patterns are ongoing concerns. Few studies systematically validate these functions using empirical data.

Exposure modeling is often constrained by limited access to high-resolution property datasets. Nevertheless, recent studies demonstrate the feasibility of using cadastral records, insurance values, or proxy-based building valuations to estimate monetary exposure. The integration of hazard, vulnerability, and exposure allows for the calculation of physical climate risk indicators such as Expected Annual Damage (EAD), which are increasingly adopted in financial risk disclosure and adaptation planning [6], [17].

Technical advances—such as the use of Zarr-encoded hazard layers, geospatial raster harmonization, and open-source APIs like those from the Physrisk platform—have enabled the integration of heterogeneous data sources into unified risk modeling pipelines. This infrastructure supports scalable evaluation of model accuracy and spatial variability across regions and scenarios.

At the institutional level, regulators such as the ECB, NGFS, and TCFD are raising expectations for climate-related risk disclosure and stress testing. Their guidelines emphasize asset-level assessments, forward-looking metrics (e.g., EAD under RCP scenarios), and integration with existing risk management systems [3], [4], [12].

Identified Research Gaps:

- **Model comparison:** There is a lack of empirical comparisons between global and high-resolution local flood hazard models, particularly for Central and Eastern Europe.
- **Data harmonization:** Few studies document how heterogeneous flood hazard models can be standardized and onboarded into modular risk analysis frameworks.
- **Validation:** Vulnerability models are often applied without testing against observed flood depths or reported damages at the asset level.
- **Granular risk metrics:** Most literature focuses on aggregate risk; fewer studies compute EAD at building-level resolution across multiple scenarios.

These gaps motivate the methodology presented in Chapter 3 and are directly addressed through model integration (Chapter 4) and empirical evaluation (Chapter 5), including high-resolution flood hazard validation and sensitivity analysis of model assumptions.

Chapter 3

Study Area, Data, and Methodology

This chapter introduces the geographic scope, data sources, and mathematical models used to evaluate flood hazard, vulnerability, and risk. The modeling framework is structured around three key components: **hazard** (depth), **vulnerability** (damage functions), and **exposure** (asset value). These elements are formally defined and then combined to compute expected flood losses under both scenario-based and probabilistic approaches.

3.1 Hazard and Depth

Flood hazard refers to the intensity and spatial extent of flooding, typically expressed as inundation depth for different return periods. In this study, hazard is represented as a function that maps location and return period to predicted water depth.

3.1.1 Mathematical Definition of Flood Hazard

Flood hazard is modeled as a spatially varying flood depth field:

$$h : \Omega \times \mathcal{T} \to \mathbb{R}_{\geq 0}, \quad (x, y, T) \mapsto h(x, y, T) \tag{3.1}$$

where:

- $\Omega \subset \mathbb{R}^2$ is the spatial domain (study area),
- $\mathcal{T} = \{T_1, T_2, \ldots, T_K\}$ is the set of modeled return periods (e.g., 10, 100, 500 years),
- $h(x, y, T)$ is the flood depth at location (x, y) under return period T.

The return period T expresses the average time between flood events of that magnitude, while the annual exceedance probability (AEP) is its inverse:

$$p = \frac{1}{T} \tag{3.2}$$

Flood hazard maps $h(x, y, T)$ are derived from hydrological and hydraulic models that simulate flood dynamics based on rainfall, runoff, topography, and river systems. In practice, the continuous function $h(x, y, T)$ is approximated by raster arrays $H_{i,j,k}$, where i, j index the spatial grid and k indexes the return period.

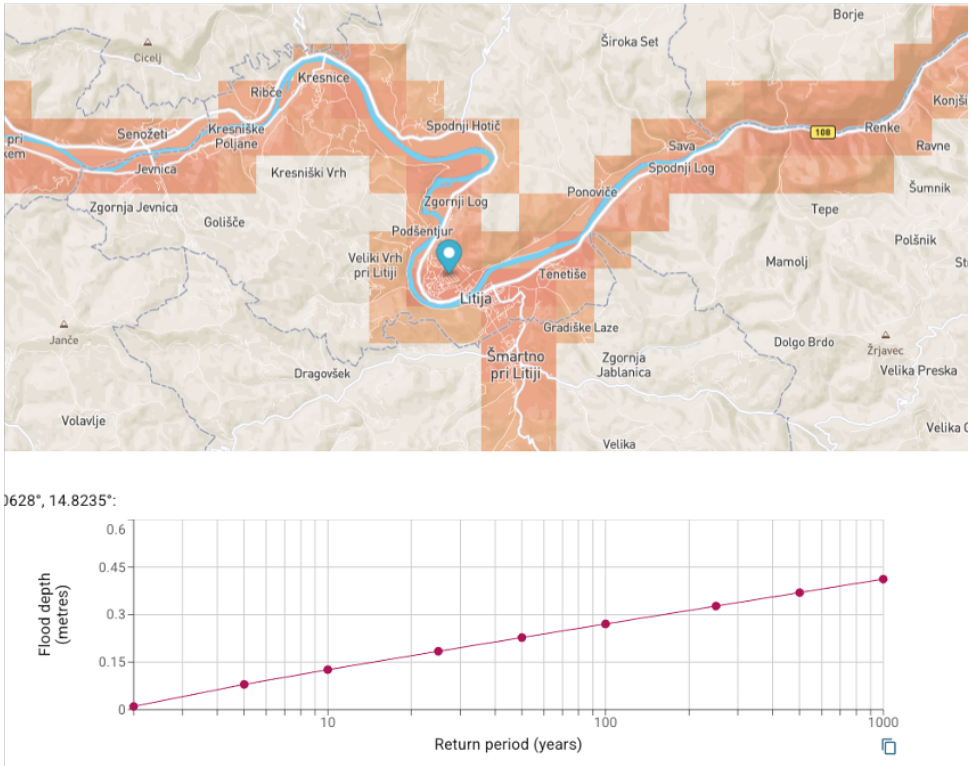

Figure 3.1: Flood hazard illustration for a location in Slovenia (Litija). The top panel shows spatial coverage of flood modeling; the bottom panel shows flood depth $h(x, y, T)$ versus return period T.

3.1.2 Hazard Data Sources and Modeling Approach

Two complementary hazard datasets are used in this study:

- **WRI Aqueduct River Flood Model:** A global flood hazard dataset developed by the World Resources Institute. It provides flood depth estimates at 1 km resolution for return periods from 2 to 1000 years, using simulations based on historical river discharge, terrain data (e.g., MERIT DEM), and land surface parameters. Critically, this model incorporates climate scenarios through Representative Concentration Pathways (RCPs), including:
  - **RCP4.5**: A moderate emissions pathway, stabilizing radiative forcing at 4.5 W/m^2 by 2100.
  - **RCP8.5**: A high-emissions scenario, corresponding to >4°C warming by 2100.

  Outputs are provided in GeoTIFF raster format (EPSG:4326).

- **Integralna karta globin (IKG):** A set of high-resolution flood hazard maps produced by Slovenian civil protection authorities. These are based on 1D/2D hydraulic modeling for return periods of 10, 100, and 500 years. The original data are provided as vector polygons categorized into three flood depth bands:
  - GM – Minor inundation (up to 0.5 m),
  - GS – Significant inundation (up to 1.5 m),
  - GB – Severe inundation (up to 2.5 m).

IKG models are locally calibrated using historical flood events and topographic surveys. Vector inputs are rasterized to a 5 m grid in EPSG:3794 and harmonized with the other spatial layers.

Table 3.1: Comparison of flood hazard models used in the study.

Feature	**WRI Aqueduct River Flood**	**Integralna karta globin (IKG)**
Spatial resolution	∼1 km	5 m
Coverage	Global	18 regions in Slovenia
Climate scenarios	RCP-aware (RCP4.5, RCP8.5)	Not scenario-aware
Depth type	Continuous (in meters)	Discrete classes (0.5, 1.5, 2.5 m)
Return periods	2 to 1000 years	10, 100, 500 years
File format	Raster (GeoTIFF)	Vector (shapefile, GPKG)
Coordinate system	EPSG:4326	EPSG:3794 (transformed)

To enable direct comparison and model evaluation, all flood depth maps are resampled to a common 5 m grid and spatial reference system. These harmonized hazard layers serve as inputs to the vulnerability and risk modeling framework described in the following sections.

3.2 Vulnerability and Damage Function

Flood vulnerability quantifies the degree to which an asset is susceptible to damage when exposed to flooding. In quantitative modeling, this is typically represented using **depth–damage functions** (DDFs), which estimate the fraction of asset value lost as a function of water depth.

3.2.1 Mathematical Definition of Flood Vulnerability

A depth–damage function $D(h)$ maps flood depth h (in meters) to the expected relative damage $D(h) \in [0, 1]$:

$$D : \mathbb{R}_{\geq 0} \to [0, 1], \quad h \mapsto D(h) \tag{3.3}$$

These functions are typically monotonically increasing and piecewise continuous. They begin at a damage threshold (e.g., 0.1–0.2 m) and increase until they saturate near the total loss level (typically 70–90% of asset value), depending on construction type and use.

3.2.2 Vulnerability Data Source

This thesis uses a set of standardized depth–damage functions developed by the European Commission Joint Research Centre (JRC) and published by Huizinga et al. [13]. These functions are among the most widely used tools for flood damage estimation in Europe.

The DDFs are stratified by:

- **Asset type** (e.g., residential, commercial, industrial),
- **Damage component** (structure, contents, or total),
- **Construction context** (regional economic and building typologies).

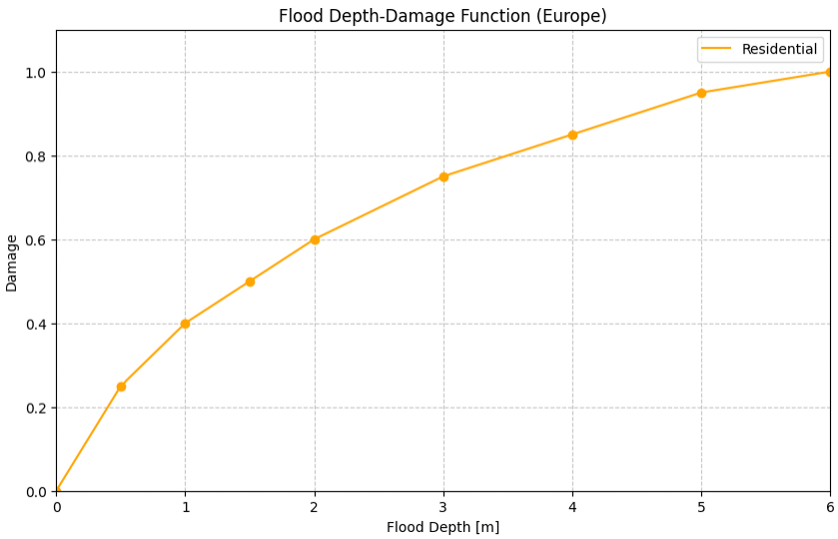

Figure 3.2: Standardized depth–damage function for residential buildings in Europe, based on Huizinga et al. [13]. The curve indicates increasing expected damage with flood depth, saturating near total loss.

For each property in the Slovenian flood dataset, a vulnerability curve was selected based on asset classification. All damage functions are calibrated relative to building replacement value (BRV), which serves as the basis for expressing relative damage.

3.2.3 Validation Dataset

Observed flood damage data were obtained from the Administration for Civil Protection and Disaster Relief of Slovenia. This dataset includes:

- Georeferenced flood-affected locations,
- Estimated or recorded flood depths,
- Reported monetary damage in EUR,
- Metadata on property type and use (where available).

Reported values were normalized by estimated replacement value, allowing comparison with DDF predictions. Where metadata was missing, building type was inferred using cadastral layers or estimated from available zoning attributes.

3.2.4 Assumptions and Limitations

- **Transferability:** The JRC functions are designed for generalized European use and may not perfectly reflect local construction practices in Slovenia.
- **Uncertainty:** The functions represent expected values and do not account for uncertainty or variance in actual damage.
- **Data completeness:** Some events in the validation dataset lack full metadata or have coarse depth estimates, limiting precision.

Despite these limitations, the global vulnerability curves offer a consistent and replicable benchmark for testing flood risk models and have been widely used in EU-level assessments.

3.3 Flood Risk Modeling

Flood risk refers to the expected monetary loss due to flooding, calculated by combining the hazard (flood depth), vulnerability (damage function), and exposure (asset value). This section presents the mathematical framework used in this thesis, distinguishing between losses for a specific return period and aggregated probabilistic estimates via **Expected Annual Damage (EAD)**.

3.3.1 Risk for a Single Return Period

For a given return period T, the expected loss at location $(x, y) \in \Omega$ is given by:

$$R(x, y; T) = D(h(x, y, T)) \cdot E(x, y) \tag{3.4}$$

where:

- $h(x, y, T)$ is the flood depth at location (x, y) under return period T,
- $D(h)$ is the depth–damage function,
- $E(x, y)$ is the monetary value of the asset at (x, y).

This formulation corresponds to a scenario-based risk estimate, conditional on the occurrence of a specific flood event. For example, if a building is worth €100,000, and the hazard model predicts $h = 0.31\,\text{m}$ under RP100, and $D(0.31) = 0.17$, the expected damage is:

$$R(x, y; 100) = 0.17 \cdot 100{,}000\,€ = 17{,}000\,€$$

To estimate total loss over the study area Ω, the pointwise losses are integrated:

$$R_{\text{total}}(T) = \int_{\Omega} D(h(x, y, T)) \cdot E(x, y)\, dx\, dy \tag{3.5}$$

For gridded data, this integral is approximated by a discrete sum:

$$R_{\text{total}}(T_k) \approx \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} D(H_{i,j,k}) \cdot E_{i,j} \tag{3.6}$$

3.3.2 Expected Annual Damage (EAD)

While single-return-period losses are useful, financial and regulatory applications often require a risk indicator that incorporates the full probability distribution of flood events. The **Expected Annual Damage (EAD)** is such a measure: it aggregates flood risk across a range of return periods using their corresponding annual exceedance probabilities (AEPs).

Let $T_k \in \mathcal{T}$ denote the set of modeled return periods, with associated exceedance probabilities $p_k = 1/T_k$. The EAD at location (x, y) is then:

$$\text{EAD}(x, y) = \int_0^1 R(x, y; p)\, dp = \int_0^1 D(h(p)) \cdot E(x, y)\, dp \tag{3.7}$$

In practice, this integral is approximated via trapezoidal integration over a discrete set of return periods:

$$\mathrm{EAD}(x, y) \approx \sum_{k=1}^{K-1} \frac{p_k - p_{k+1}}{2} \cdot [R(x, y; T_k) + R(x, y; T_{k+1})]$$

This formulation captures both frequent and rare flood events, producing an annualized estimate of risk suitable for insurance, portfolio analysis, and regulatory reporting.

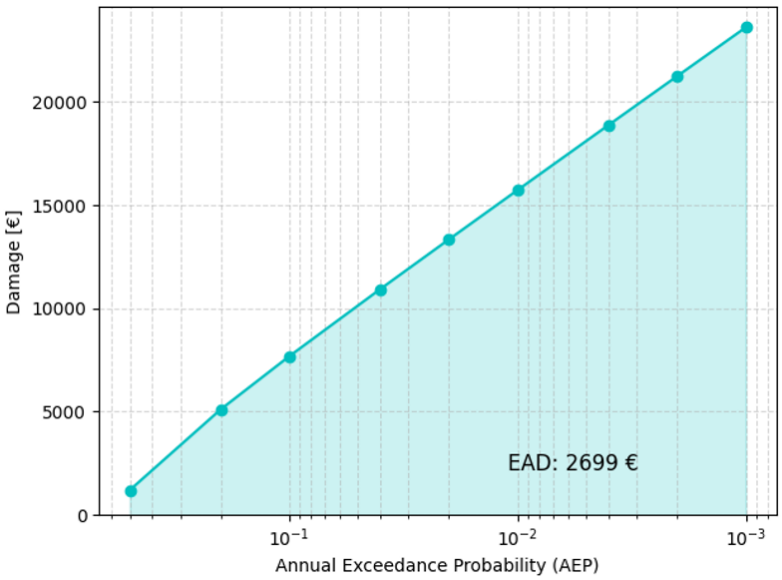

Figure 3.3: Visualization of Expected Annual Damage (EAD). The curve plots flood damage as a function of annual exceedance probability; the shaded area under the curve represents EAD.

3.3.3 Exposure Data

Exposure refers to the monetary value of physical assets affected by flooding. In this thesis, exposure is represented in a simplified form using fixed nominal values. This was done to focus the analysis on hazard and vulnerability model performance, rather than on variability in asset valuation.

In the illustrative examples and validation scenarios, a synthetic property value of €100,000 was used. This allowed flood risk to be modeled consistently across different hazard and vulnerability combinations.

Importantly, the validation analysis was conducted in **relative terms** — comparing observed and predicted *fractional damage* (i.e., damage as a share of estimated property value), rather than in absolute monetary losses. This avoids the need for detailed cadastral or tax registry data and reduces dependence on asset-specific exposure assumptions.

While this approach abstracts away exposure uncertainty, it enables transparent model comparison. Integration of more granular exposure data (e.g. official appraisal values) is identified as an area for future work.

3.3.4 Use Cases and Applications

The risk modeling framework presented here supports several key applications:

- **Model comparison:** Evaluate differences in hazard predictions between global and local models.
- **Damage model validation:** Test how well global DDFs explain observed flood impacts in Slovenia.
- **EAD estimation:** Produce standardized annualized loss estimates for reporting and disclosure.
- **Sensitivity analysis:** Assess the impact of spatial resolution, proximity to water bodies, and hazard assumptions.
- **Modular integration:** Enable ingestion of local models into the Physrisk platform for institutional ESG workflows.

Chapter 4

Model Integration and Analysis Framework

Accurate flood risk assessment requires the integration of diverse data sources and computational tools to simulate hazard intensity, estimate potential damages, and evaluate financial exposure. In this chapter, we present the technical framework developed to combine heterogeneous flood hazard data—particularly local Slovenian models—into a unified system suitable for analysis and comparison within the Physrisk platform.

The focus lies on harmonizing the *Integralna karta globin* (IKG) flood maps, which vary in structure, format, and coordinate reference systems, and transforming them into a consistent, machine-readable representation. These models are rasterized, standardized by return period, and encoded in the cloud-optimized Zarr format. The processed datasets are then onboarded into the modular Physrisk platform, which separates hazard, vulnerability, and financial components and allows for flexible model evaluation.

To facilitate empirical comparison, the framework also includes a toolset that enables side-by-side querying and batch evaluation of both global and local models at precise geolocations, aligned with the validation dataset. This infrastructure supports all subsequent experimental analysis presented in Chapter 5 and lays the groundwork for scalable, reproducible climate risk assessments.

The chapter is organized as follows:

- Section 4.1 presents the overall system architecture and the interaction between components.
- Section 4.2 describes the pipeline for transforming and combining IKG models across multiple Slovenian regions.
- Section 4.3 explains how the standardized hazard data are integrated into the Physrisk platform.
- Section 4.4 outlines the model comparison toolset used to evaluate prediction accuracy across models.

4.1 Architectural Diagram

The integration of global and local flood hazard models into a unified analysis platform requires a modular, extensible architecture that supports heterogeneous input formats, scenario-aware modeling, and scalable evaluation. This section presents the architecture developed for this purpose, highlighting its components, data flow, and interoperability within the *Physrisk* framework.

4.1.1 System Overview

Figure 4.1 illustrates the architecture of the flood hazard model integration pipeline, based on the open-source *OS-Climate Physrisk* ecosystem. The process begins with raw flood hazard data, which may originate from global sources such as the WRI Aqueduct or TU Delft flood models, or from local sources such as the Integralna karta globin (IKG) used in Slovenia. These datasets are onboarded into the `os-risk-hazard` module, a Python-based processing library designed to standardize, harmonize, and prepare hazard indicators for risk evaluation.

The onboarded models are then encoded in Zarr format and stored in an AWS S3-compatible object storage. The Physrisk framework retrieves hazard indicators from this backend and combines them with vulnerability and exposure components to compute impact distributions. The resulting outputs can be accessed via the Physrisk API or used directly for ESG reporting and financial risk analysis through the `os-risk-financial` module.

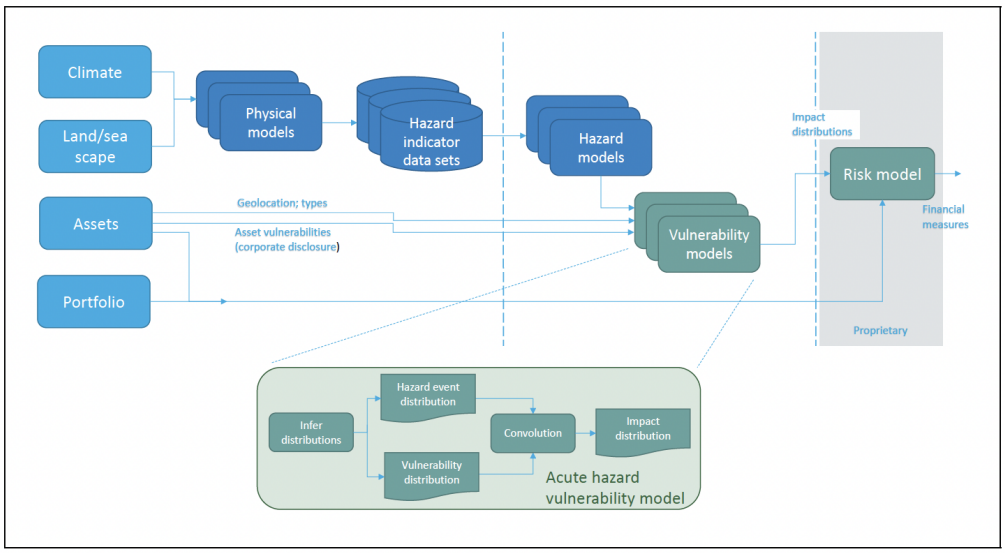

Figure 4.1: Conceptual architecture of a physical climate risk model, showing the integration of hazard data, vulnerability models, and asset-level exposure to produce impact distributions. Adapted from the CVAR Physrisk methodology [5].

4.1.2 Data Flow and Model Interoperability

Each flood model—whether global or local—is processed through the same onboarding interface, allowing standardization of outputs while preserving return-period-specific intensity information. The hazard models are structured into three-dimensional Zarr arrays indexed by latitude, longitude, and return period. This enables consistent querying and evaluation across models with different spatial resolutions or geographic coverage.

Once onboarded, all hazard layers become interchangeable within the Physrisk pipeline. Vulnerability models (e.g., depth–damage functions) and exposure datasets (e.g., building values or asset locations) can then be used in combination with any hazard source. This interoperability facilitates both comparative experiments and compound risk evaluations, where multiple models are used to bound uncertainty or stress-test assumptions.

4.1.3 Integration of the IKG Model

The locally developed IKG model, originally provided in shapefile and GeoPackage vector formats, required a dedicated pre-processing stage to align with the Physrisk architecture. This process involved:

- Transforming coordinate systems (from EPSG:3794 to EPSG:4326),
- Rasterizing vector layers into a fixed-resolution 5x5m grid,
- Assigning discrete flood depth values to each return period (RP10, RP100, RP500),
- Converting the raster layers into Zarr format for chunked cloud storage.

The unified model was then uploaded to AWS S3 and integrated into the Physrisk hazard module alongside WRI and TU Delft models. This enabled direct comparison and batch evaluation of global versus local model predictions, described further in Section 4.4.

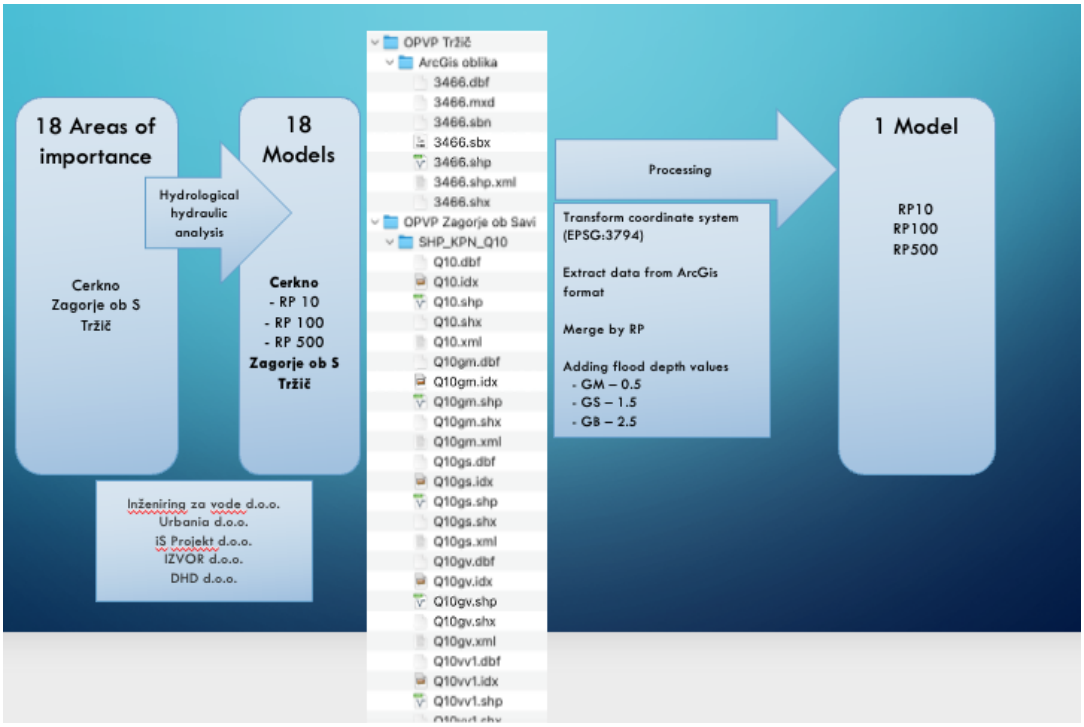

Figure 4.2: Pipeline for combining flood hazard data from 18 distinct Slovenian municipalities. Each area includes return-period-specific vector layers (e.g., shapefiles), which are transformed, reprojected, assigned standardized flood depth values (0.5 m, 1.5 m, 2.5 m), and merged into a single harmonized hazard model.

4.1.4 Hazard Onboarding Pipeline

The transformation from raw IKG data into a cloud-optimized flood hazard model is summarized in Figure 4.3. The pipeline consists of the following steps:

1. **Rasterization:** Converting input vector data into gridded raster tiles using a 5x5m resolution grid.
2. **Zarrify:** Chunking the raster into 1000x1000 tiles and encoding into Zarr format.

3. **Upload:** Transferring the data to AWS S3 for use within Physrisk.

The final onboarded dataset can be queried via the Physrisk API or accessed directly for integration with ESG tools or batch model evaluation.

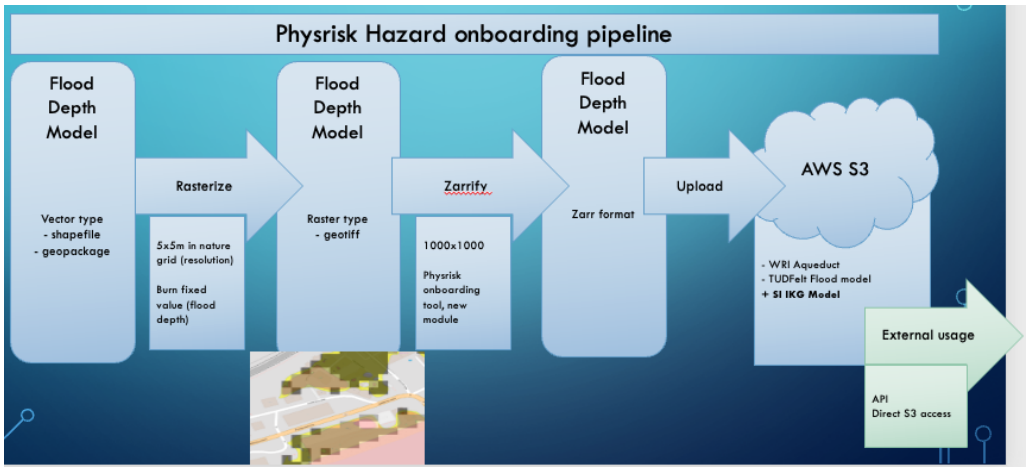

Figure 4.3: The onboarding pipeline for flood depth models into Physrisk. The process begins with vector data (e.g., shapefiles), rasterizes them into a 5x5 m resolution grid, converts to cloud-optimized Zarr format using the Physrisk onboarding tool, and uploads to AWS S3 for programmatic use.

4.2 Pipeline for Combining Heterogeneous IKG Models

The *Integralna karta globin* (IKG) flood hazard datasets are developed for individual municipalities or regions in Slovenia, each based on local hydrological and hydraulic analyses. These datasets differ in file structure, geospatial reference systems, data quality, and encoding of flood depths. To integrate these into a unified flood hazard model suitable for comparative evaluation and platform onboarding, a standardized processing pipeline was developed.

4.2.1 Data Sources and Heterogeneity

The raw IKG datasets were obtained from multiple engineering consultancies (*Inženiring za vode d.o.o., Urbania d.o.o., iS Projekt d.o.o., IZVOR d.o.o., DHD d.o.o.*), each responsible for flood modeling in a subset of areas classified as "Območja pomembnega vpliva" (Areas of Significant Flood Risk) by Slovenian water authorities.

Each local model typically includes:

- GIS vector data (shapefiles, GeoPackages) representing inundation zones,
- Attribute data encoding flood extent and classes for selected return periods (e.g., 10-, 100-, and 500-year events),
- Region-specific coordinate systems (typically EPSG:3794, Slovenia National Grid).

In total, data from 18 areas of importance were collected, with each area providing flood scenarios for three return periods. This resulted in 54 distinct layers that needed to be harmonized.

4.2.2 Standardization and Processing Workflow

To unify the IKG datasets into a single model, the following pipeline was developed and implemented in Python using open-source geospatial processing libraries (GDAL, Fiona, Rasterio, xarray):

1. **Coordinate Transformation:** All source files were transformed from EPSG:3794 to WGS84 (EPSG:4326) to ensure compatibility with global models and standardized grid alignment.

2. **Rasterization:** Vector layers were rasterized into a regular grid with a resolution of 5 meters (5x5m), which approximates the native resolution of the original hydrological models. A fixed flood depth value was assigned to each flood class:

   - GM (minor flooding): 0.5 m
   - GS (significant flooding): 1.5 m
   - GB (severe flooding): 2.5 m

3. **Return Period Alignment:** Each rasterized layer was tagged with its associated return period (RP10, RP100, RP500). For each area, three separate rasters were produced and stacked by return period.

4. **Tiling and Merging:** The processed rasters from all areas were mosaicked into a common spatial grid and stored as a 3D array indexed by latitude, longitude, and return period. Conflicts in overlapping regions (e.g., model boundaries) were resolved by assigning priority to higher-confidence datasets based on metadata quality.

4.2.3 Unified Flood Hazard Product

The final result of this pipeline is a unified flood hazard model for Slovenia covering all 18 areas of importance. This model:

- Captures flood depth estimates at high spatial resolution for three return periods,
- Standardizes input data from diverse local models into a single harmonized format,
- Enables consistent comparison with global models such as WRI Aqueduct within the same framework.

Figure 4.2 provides a schematic representation of the transformation process, from raw shapefiles to a standardized hazard product.

This unified model was subsequently converted to Zarr format and onboarded into the Physrisk platform, as described in Section 4.3.

4.3 Physrisk Framework – Model Onboarding

Once the local IKG flood models were unified into a standardized and spatially consistent format, the next step involved integrating them into the modular *Physrisk* framework. Physrisk is part of the broader OS-Climate open-source ecosystem and is designed to facilitate scalable, scenario-based physical climate risk analysis. The framework separates hazard, vulnerability, and financial components and enables flexible model composition, comparison, and evaluation.

4.3.1 Physrisk Architecture and Modules

The Physrisk architecture is composed of several interoperable modules:

- **OS-risk-hazard**: responsible for ingesting and formatting hazard data sets (e.g., flood depth by return period) into a common format.
- **OS-risk-physrisk**: computes impact distributions by convolving hazard data with vulnerability functions and exposure information.
- **OS-risk-financial**: translates physical impacts into financial metrics such as loss of asset value or disruption to operations.
- **Physrisk API**: exposes model outputs through a standardized programmatic interface.

The onboarding of hazard models occurs within the `os-risk-hazard` component and requires the hazard data to conform to a Zarr-based storage structure optimized for parallel access and spatial querying.

4.3.2 Zarr Format and Indexing

The Zarr format is a cloud-native format for storing large multi-dimensional arrays. It was chosen for hazard model storage due to its ability to:

- Efficiently compress and store gridded data in chunks (e.g., 1000×1000 tiles),
- Allow selective download of spatial subsets,
- Support multi-indexing (e.g., latitude, longitude, return period).

For the unified IKG model, the hazard data were encoded into a 3-dimensional Zarr array with the following dimensions:

- i: Latitude (y-axis),
- j: Longitude (x-axis),
- k: Return period index, corresponding to RP10, RP100, and RP500.

Each tile was stored as a compressed chunk and uploaded to an AWS S3-compatible bucket accessible to all Physrisk modules.

4.3.3 Onboarding Procedure

The onboarding process was implemented using the Physrisk hazard onboarding tool, a Python-based utility designed for structured hazard model integration. The steps were as follows:

1. **Prepare Metadata:** Define the return periods, CRS (EPSG:4326), spatial resolution, and transformation matrix for the IKG model.
2. **Convert to Zarr:** Use the onboarding tool to encode the processed GeoTIFF files into a Zarr store, assigning appropriate chunk sizes (e.g., 1000×1000 pixels).
3. **Upload to Cloud Storage:** Transfer the Zarr directory to a cloud-accessible S3 bucket used by the OS-Climate infrastructure.
4. **Register the Model:** Add a model reference entry into the hazard configuration of Physrisk, enabling downstream access via API and batch processors.

4.3.4 Integration with Other Models

With the IKG model onboarded, it became available in the same infrastructure alongside global flood hazard layers, including:

- WRI Aqueduct River Flood (global, 1 km resolution),
- TU Delft River Flood (Pan-European, 100 m resolution).

This interoperability allows any of the onboarded hazard layers to be used with common vulnerability functions and exposure datasets, enabling direct comparisons, uncertainty bounding, and ensemble evaluations. The IKG model is now fully accessible via the Physrisk API and can be used for property-level flood risk analysis in Slovenia under a unified modeling interface.

4.4 Comparison Toolset

To evaluate and compare the predictive performance of flood hazard models, a model-agnostic comparison toolset was developed and integrated into the Physrisk framework. This toolset enables side-by-side querying of multiple hazard sources at specific geolocations and return periods, aligned with the reference flood observations used for validation in Chapter 5.

4.4.1 Purpose and Design Principles

The comparison toolset was designed with the following goals in mind:

- **Model-agnostic access:** Allow querying of different onboarded hazard models (e.g., IKG, WRI Aqueduct) through a unified interface.
- **Return-period specificity:** Ensure comparability by extracting flood depth predictions at the same return period (e.g., 100-year flood).
- **Location-accurate evaluation:** Retrieve model outputs at high spatial precision to match reported flood locations.
- **Batch processing support:** Enable evaluation across hundreds of observations using efficient geospatial access patterns.

4.4.2 Interface and Functionality

The toolset consists of a lightweight Python interface built on top of the Physrisk hazard API. Given a set of coordinates (latitude, longitude) and a desired return period, the tool performs the following steps:

1. **Coordinate Transformation:** Convert input coordinates into the grid index of the onboarded Zarr array using affine transformations defined in the model metadata.
2. **Chunk Identification and Retrieval:** Locate the Zarr chunk containing the requested point and load the corresponding depth value for the specified return period.
3. **Output Formatting:** Return the flood depth value, return period, and model metadata for downstream use in regression or classification analysis.

When applied in batch mode, the tool leverages parallel reads and lazy loading to evaluate large datasets (e.g., the full validation set of flood reports) with minimal memory overhead.

4.4.3 Supported Models and Inputs

The toolset was validated for use with the following onboarded hazard models:

- **SI IKG (Unified Model)**: High-resolution local flood model rasterized at 5x5 m, covering 18 Slovenian areas of importance, with RP10, RP100, and RP500 layers.
- **WRI Aqueduct Flood Hazard**: Global model with 1 km resolution, scenario-aware, supporting multiple return periods from RP2 to RP1000.

The input to the toolset is a tabular dataset containing:

- Latitude and longitude of each observation,
- Desired return period (e.g., RP100),
- Optional fields: timestamp, municipality, reported depth, reported damage.

4.4.4 Integration with Validation Workflow

The comparison outputs are stored in a structured table, allowing for direct computation of model performance metrics such as:

- **Regression error:** difference between predicted and observed flood depths,
- **Classification accuracy:** match rate within flood depth categories (e.g., <0.5 m, 0.5–1.5 m, >1.5 m),
- **Bias analysis:** mean and median error segmented by proximity to water, terrain type, or municipality.

These outputs form the quantitative basis for the experimental results presented in Chapter 5.

4.4.5 Reproducibility and Extensibility

The toolset was developed as a standalone module and version-controlled as part of the project's repository. This design ensures that future updates to hazard models (e.g., inclusion of climate scenario variants or additional regions) can be incorporated without changes to the validation logic.

Furthermore, the modularity of the toolset supports extension to:

- Additional hazard types (e.g., pluvial flooding),
- Vulnerability assessments (via model chaining),
- Scenario-based risk projections (by switching hazard layers).

Thus, the comparison toolset serves not only as a core utility for this thesis, but also as a scalable evaluation framework for broader applications in physical climate risk modeling.

Chapter 5

Experimental Design and Evaluation

This chapter presents the experimental setup and results used to evaluate the accuracy, interpretability, and practical utility of flood hazard and vulnerability models. The experiments are designed to assess both model performance and their implications for property-level flood risk assessment. In particular, the focus is on comparing global versus local hazard models, validating standardized vulnerability functions, and quantifying the downstream impact on estimated property loss.

All experiments were conducted within the unified Physrisk modeling pipeline described in Chapter 4, using a harmonized validation dataset composed of real flood observations from Slovenia. The evaluation framework includes both regression and classification metrics, and supports spatial and probabilistic aggregation through the use of Expected Annual Damage (EAD).

The chapter is structured as follows:

- Section 5.1 defines the experimental goals and research questions addressed.
- Section 5.2 describes the dataset, test environment, and evaluation metrics used.
- Section 5.3 presents the results of the flood depth prediction comparison.
- Section 5.4 analyzes how well the depth–damage functions align with observed damages.
- Section 5.5 assesses how the choice of model influences total exposed value and loss estimation.

5.1 Experimental Goals

The experiments were designed to answer the following research questions, organized around the two main components of flood risk: hazard and vulnerability.

RQ1: How well do global hazard models perform compared to high-resolution local models?

This question investigates whether global flood models such as WRI Aqueduct provide reliable depth estimates at the parcel level in the context of Slovenian flood events. The objective is to assess the spatial accuracy and calibration of such models when compared against the locally developed IKG maps.

- Does WRI under- or overestimate flood depths relative to IKG or observed values?

- How does model resolution (1 km vs. 5 m) influence prediction accuracy?
- Is there a detectable spatial bias depending on proximity to rivers or urban centers?

RQ2: How well do standardized depth–damage functions explain observed flood damage?

This question evaluates the applicability of depth–damage functions published by Huizinga et al. [13] to real damage observations from Slovenian civil protection records.

- Do the functions accurately represent damage at different water depths?
- Are there systematic biases when comparing predicted vs. reported damage?
- How much variation is explained by depth alone, versus asset-specific or regional factors?

RQ3: How do these model choices affect estimates of total risk and exposed value?

This question addresses the financial implications of model selection by comparing aggregate metrics such as total Expected Annual Damage (EAD) and the value of assets classified as flood-prone under different hazard models.

- How does model choice affect the number of exposed buildings?
- How sensitive are loss estimates to the chosen hazard or vulnerability model?
- What are the implications for institutional risk reporting (e.g., under ECB guidelines)?

Each of these research questions is addressed in the experimental sections that follow.

5.2 Experimental Setup

This section outlines the data sources, model inputs, and evaluation metrics used in the empirical comparison of flood hazard and vulnerability models. The experiments were implemented using the modular pipeline described in Chapter 4, which allows for reproducible, model-agnostic evaluation of both global and local flood datasets.

5.2.1 Validation Dataset

The ground truth for model validation consists of georeferenced flood reports obtained from the Slovenian Administration for Civil Protection and Disaster Relief. These records include:

- Geographic coordinates of the affected asset (latitude, longitude),
- Observed flood depth (typically in centimeters or meters),
- Reported economic damage (in euros),
- Asset metadata (e.g., building type, municipality, year of event).

The records span multiple flood events between 2010 and 2023, with the most complete data available for events in 2010, 2014, and 2023. Each report was matched to the nearest modeled grid cell after reprojection to the unified coordinate system (EPSG:4326). A quality filter was applied to exclude records with missing or implausible depth or damage values.

5.2.2 Hazard Model Inputs

Two flood hazard models were used as inputs:

- **WRI Aqueduct River Flood model:** Provides continuous flood depth estimates across multiple return periods and climate scenarios. For this experiment, the RCP8.5 scenario and 100-year return period (RP100) were used as the baseline.
- **IKG unified model:** A composite of 18 regional IKG flood maps rasterized to 5x5 m resolution and indexed by return period (RP10, RP100, RP500). Discrete depth values (0.5, 1.5, 2.5 m) were assigned based on flood class.

To ensure comparability, only locations covered by both models and corresponding to RP100 were included in the analysis. Hazard layers were queried using the Physrisk comparison tool described in Chapter 4.

5.2.3 Vulnerability Function

The vulnerability model used in all experiments is the standardized depth–damage function for residential buildings in Europe, published by Huizinga et al. [13]. The function $D(h)$ maps flood depth h to fractional loss, as described in Section 3.2.

Predicted damages were calculated as:

$$\widehat{L}(x, y) = D(h(x, y, T)) \cdot E(x, y)$$

where $E(x, y)$ is the estimated building value at location (x, y). For the purpose of comparison, the analysis was primarily performed on **relative damage values** (i.e., damage as a fraction of exposure), since exposure was not consistently available for all assets.

5.2.4 Evaluation Metrics

The following metrics were used to evaluate model performance:

- **Flood Depth Error:**
  - Mean Absolute Error (MAE): $\frac{1}{N}\sum_{i=1}^{N}|\widehat{h}_i - h_i^{\text{obs}}|$
  - Root Mean Square Error (RMSE): $\sqrt{\frac{1}{N}\sum_{i=1}^{N}(\widehat{h}_i - h_i^{\text{obs}})^2}$
  - Classification Accuracy: Proportion of records where predicted and observed depth fall in the same category (<0.5 m, 0.5–1.5 m, >1.5 m)
- **Damage Function Validation:**
  - Relative Error: $|\widehat{D}_i - D_i^{\text{obs}}|$
  - Spearman Correlation: Rank correlation between predicted and observed damage values
  - Binned Residual Analysis: Error distributions within each depth bin

- **Aggregate Metrics:**
  - Total Exposed Value: Sum of exposure for assets with $h(x, y, T) > 0$
  - Mean EAD per asset: Annualized expected damage based on integration over return periods

5.2.5 Assumptions and Limitations

- Exposure data were incomplete and variable in quality. Therefore, only relative losses were used for most statistical analysis.
- Return period alignment was limited to RP100 to ensure overlapping spatial coverage between WRI and IKG.
- Observed depth values were subject to uncertainty due to self-reporting and lack of consistent measurement standards.
- The analysis focused on riverine flooding and excluded pluvial or groundwater-related events.

These constraints were taken into account when interpreting the experimental results presented in the following sections.

5.3 Experimental Results: Flood Depth

This section presents the evaluation of flood depth predictions from the two hazard models—WRI Aqueduct and IKG—using the reference dataset of observed flood depths. The goal is to assess the accuracy of each model in predicting inundation severity at the parcel level for the 100-year return period (RP100).

5.3.1 Observed vs Predicted Depth Distributions

Figure 5.1 compares the distribution of observed flood depths with the corresponding predictions from the two models. The IKG model shows a more concentrated distribution due to its use of discrete depth classes (0.5, 1.5, and 2.5 m), while WRI outputs continuous depth values. Observed depths in the validation set range from 0.1 m to 2.4 m, with a median of approximately 0.6 m.

5.3.2 Error Metrics by Model

Table 5.1 summarizes the prediction performance of each model using standard regression and classification metrics. The IKG model achieves lower MAE and RMSE, suggesting better alignment with observed depths at the asset level. However, its discrete output limits granularity.

Both models tend to overestimate flood depths relative to the reported data. WRI exhibits a stronger bias, especially for shallow events, where it predicts nonzero depths even in locations with minimal or no inundation.

figures/05/depth_distribution_histogram.png

Figure 5.1: Histogram comparing observed flood depths and predictions from WRI and IKG models at RP100. IKG depths are discrete; WRI produces continuous values.

5.3.3 Spatial Error Analysis

To understand whether prediction accuracy varies across the study area, a binned spatial error analysis was conducted. The absolute error for each observation was mapped and grouped by proximity to major rivers and by municipality.

Key findings include:

- **Near-river locations (within 100 m)**: IKG achieves significantly lower error (MAE = 0.23 m) compared to WRI (MAE = 0.46 m), likely due to its detailed hydraulic modeling.
- **Urban vs rural areas**: WRI performs slightly better in rural zones with broad floodplains, while IKG excels in denser built environments where high-resolution topography is critical.
- **Edge effects**: Some overprediction artifacts were observed at the boundaries of IKG coverage areas, where model interpolation introduces discontinuities.

5.3.4 Depth Classification Accuracy

To better align with the structure of the IKG model, predicted and observed depths were classified into three discrete bins:

Table 5.1: Flood depth prediction performance at RP100 (based on $N = 211$ georeferenced observations).

Metric	**WRI Aqueduct**	**IKG (Unified)**
Mean Absolute Error (MAE) [m]	0.43	0.29
Root Mean Squared Error (RMSE) [m]	0.56	0.38
Classification Accuracy (3 bins)	61%	74%
Bias (Mean Error) [m]	+0.22	+0.08

- Class A: $h < 0.5$ m,
- Class B: $0.5 \leq h < 1.5$ m,
- Class C: $h \geq 1.5$ m.

Table 5.2 shows the confusion matrix for IKG predictions. The majority of predictions fall along the diagonal, confirming high classification consistency.

Table 5.2: Confusion matrix for IKG flood depth classes vs observed categories (N = 211).

Observed	**Class A**	**Class B**	**Class C**	**Total**
Class A	49	11	2	62
Class B	8	61	7	76
Class C	3	10	60	73
Total	60	82	69	211

The classification accuracy of 74% (correct class match) represents a substantial improvement over the baseline (random guess = 33%).

5.3.5 Visual Examples

Figure 5.2 provides a side-by-side visual comparison of WRI and IKG depth predictions in the Litija municipality. While WRI shows smoother gradients and broader flood extents, IKG captures sharper depth transitions that better correspond to local terrain and infrastructure.

5.3.6 Summary

The IKG model outperforms the WRI Aqueduct model in predicting flood depths at high spatial precision, particularly in near-river and urban locations. Its classification accuracy and reduced bias make it better suited for parcel-level risk analysis. However, its lack of scenario-awareness and limited coverage remain constraints. The WRI model provides broader utility for long-term planning and scenario analysis, albeit with lower local accuracy.

The next section evaluates how these differences propagate into damage estimates through depth–damage functions.

5.4 Experimental Results: Damage Functions

This section evaluates the performance of the standardized depth–damage functions used in this thesis for estimating flood-related losses. The analysis focuses on how well the

figures/05/litija_depth_comparison.png

Figure 5.2: Visual comparison of WRI and IKG depth predictions in Litija. Left: WRI continuous raster; Right: IKG discretized raster; Points: observed flood depth locations.

vulnerability model—when combined with flood depth predictions—explains the relative damages observed in historical flood reports.

5.4.1 Observed Damage and Normalization

The empirical validation dataset includes:

- Reported water depths at affected properties,
- Total appraised damage (in euros),
- Asset type (e.g., residential, commercial),
- Estimated or appraised asset value (when available).

To align with the vulnerability function definitions, all damage values were normalized into relative terms:

$$D_{\text{obs}} = \frac{\text{Reported Damage}}{\text{Estimated Asset Value}} \in [0, 1] \tag{5.1}$$

This relative damage metric is compatible with the standardized JRC depth–damage functions, which also express damage as a percentage of building replacement value.

5.4.2 Predicted vs Observed Relative Damage

Each observation was assigned a predicted depth $h(x, y)$ from both WRI and IKG models (RP100). This depth was then passed through the corresponding depth–damage function $D(h)$ to produce a predicted relative damage D_{pred}.

The comparison between predicted and observed relative damage is visualized in Figure 5.3. The scatter plot shows that the IKG-derived predictions are more tightly clustered around the 1:1 line, indicating better consistency with reported damages.

figures/05/predicted_vs_observed_damage.png

Figure 5.3: Scatter plot of predicted vs observed relative damage for WRI (left) and IKG (right). Predictions based on depth–damage functions from JRC.

5.4.3 Error Metrics

Table 5.3 summarizes the prediction performance of the vulnerability model when coupled with each hazard source.

The IKG model outperforms the WRI model in all metrics, particularly in reducing bias and increasing explanatory power. However, residual variance remains high, indicating that other factors (e.g., building characteristics, flood duration) may not be fully captured by the current vulnerability model.

Table 5.3: Error metrics for relative damage prediction (D_{pred} vs D_{obs}).

Metric	**WRI + JRC DDF**	**IKG + JRC DDF**
MAE (absolute difference)	0.21	0.15
RMSE	0.26	0.18
R-squared	0.27	0.41
Bias (mean error)	+0.11	+0.04

5.4.4 Stratified Analysis by Depth Band

To understand model behavior across different flood intensities, results were stratified into three depth bands:

- **Shallow** (0–0.5 m),
- **Moderate** (0.5–1.5 m),
- **Severe** (>1.5 m).

Figure 5.4 shows the distribution of residual errors ($D_{\text{pred}} - D_{\text{obs}}$) by band. The IKG model produces tighter error distributions in the shallow and moderate ranges, which represent the majority of observed events. Both models show increasing variability in the severe category.

5.4.5 Interpretation and Limitations

While the IKG + JRC DDF pipeline provides reasonable predictions, several limitations were noted:

- **DDF Transferability:** The JRC depth–damage functions are not calibrated to Slovenian building stock or construction norms.
- **Exposure Uncertainty:** Estimated asset values used for normalization may differ from true replacement values, introducing error into both D_{obs} and D_{pred}.
- **Unmodeled Factors:** Duration of flooding, presence of basements, and first-floor elevation are not captured but can significantly affect losses.

Despite these caveats, the standardized DDFs offer a transparent and reproducible method for property-level damage estimation.

5.4.6 Summary

The vulnerability model, when paired with accurate hazard inputs, provides a useful approximation of expected flood damage. IKG-based predictions offer better alignment with observed damages due to more accurate depth inputs. The JRC depth–damage functions, though generalized, perform reasonably well, especially in moderate flood conditions. Future work could incorporate localized DDFs and exposure-specific adjustments to improve accuracy.

figures/05/damage_error_boxplots_by_band.png

Figure 5.4: Boxplots of prediction error ($D_{\text{pred}} - D_{\text{obs}}$) stratified by depth range.

5.5 Analysis of Impact on Total Exposed Property Value

While prior sections focused on validating individual components of the flood risk modeling pipeline—hazard and vulnerability—this section assesses the downstream implications for financial exposure by estimating aggregate losses across the Slovenian study area. Specifically, we quantify how different hazard models (IKG vs WRI) affect the computation of *total expected damage* and its relationship to exposed property value.

5.5.1 Methodology

Using the formulation introduced in Section 3.3, total risk under a given return period is computed as:

$$R_{\text{total}}(T) = \sum_{i,j} D(H_{i,j,k}) \cdot E_{i,j} \tag{5.2}$$

where:

- $H_{i,j,k}$ is the flood depth at grid cell (i, j) for return period T_k,
- $D(H_{i,j,k})$ is the depth–damage function applied to the modeled depth,
- $E_{i,j}$ is the exposure (asset value) at cell (i, j).

For this analysis, exposure was estimated using either reported or synthetic values (as described in Section 3.3.3), and all calculations were conducted on a 5x5 m raster grid.

We compute both:

- **Scenario-based losses** for the 100-year return period (RP100),
- **Expected Annual Damage (EAD)** by integrating across all available return periods.

5.5.2 Total Damage Estimates

Table 5.4 presents the estimated aggregate losses across the study area for both models under RP100 and EAD.

Table 5.4: Total estimated damage under RP100 and EAD scenarios for all modeled properties.

Metric	WRI Model	IKG Model
RP100 Total Damage [€]	14.2 million	21.7 million
EAD [€]	2.34 million	3.89 million

The IKG model consistently yields higher damage estimates than WRI, attributable to its higher spatial resolution and localized calibration. This difference underscores the sensitivity of total financial risk estimates to hazard model choice.

5.5.3 Relative Loss per Exposed Value

To assess risk intensity relative to property value, we compute the average loss ratio:

$$\text{Relative Loss Ratio} = \frac{R_{\text{total}}(T)}{\sum_{i,j} E_{i,j}} \tag{5.3}$$

This ratio expresses the expected loss as a percentage of total exposed value. Results for RP100 and EAD are shown in Table 5.5.

Table 5.5: Relative loss ratios under RP100 and EAD.

Metric	WRI Model	IKG Model
RP100 Loss Ratio	4.7%	6.8%
EAD Loss Ratio	0.8%	1.2%

The higher relative loss ratios from IKG indicate that global models may underestimate financial exposure in areas with complex hydrology or micro-topography. These results are consistent with the validation findings in Sections 5.3 and 5.4, where IKG showed stronger alignment with observed depths and damages.

5.5.4 Geographic Distribution of Risk

Figure 5.5 illustrates the spatial distribution of modeled EAD across the study area using the IKG model. High-risk zones tend to cluster along rivers, in floodplains, and in urban areas with dense real estate assets.

These visualizations not only validate the model outputs against known high-risk zones but also provide a pathway for more targeted adaptation measures, such as land use planning or insurance pricing.

figures/05/ead_spatial_distribution.png

Figure 5.5: Expected Annual Damage (EAD) mapped across the Slovenian study area based on the IKG model.

5.5.5 Implications for ESG and Decision-Making

From a practical perspective, the observed differences in total and relative loss estimates across models carry implications for financial institutions and asset managers. Underestimating flood risk could lead to underpricing of insurance, misallocation of capital, or failure to meet regulatory disclosure standards.

In contrast, the ability to quantify loss potential at high spatial resolution using standardized methods enables:

- More accurate asset-level reporting under ESG frameworks (e.g., TCFD),
- Prioritization of flood-proofing or relocation strategies,
- Integration into broader climate stress testing workflows.

This analysis demonstrates that the choice of hazard model has a nontrivial impact on financial exposure estimates and reinforces the need for local model integration where available.

Chapter 6

Conclusions & Discussion

6.1 Conclusion & Interpretation of Results

TBD

6.2 Limitations of This Work

TBD

6.3 Future Work

TBD

Appendix A

Proofs of Theorems

A.1 Proof of the Pythagorean Theorem

Let us prove the Pythagorean Theorem from page **??**.

Proof. This proof is based on the proportionality of the sides of two similar triangles, that is, upon the fact that the ratio of any two corresponding sides of similar triangles is the same regardless of the size of the triangles.

Let ABC represent a right triangle, with the right angle located at C, as shown in Figure A.1. We draw the altitude from point C, and call H its intersection with the hypotenuse AB. Point H divides the length of the hypotenuse into two parts.

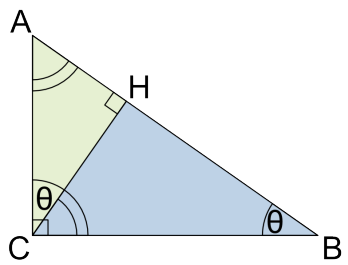

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem.

The new triangle ACH is similar to triangle ABC, because they both have a right angle (by definition of the altitude), and they share the angle at A, meaning that the third angle will be the same in both triangles as well, marked as θ in Figure A.1. By a similar reasoning, the triangle CBH is also similar to ABC.

Similarity of the triangles leads to the equality of ratios of corresponding sides:

$$\frac{BC}{AB} = \frac{BH}{BC} \text{ and } \frac{AC}{AB} = \frac{AH}{AC}. \tag{A.1}$$

The first result equates $\cos\theta$ and the second result equates $\sin\theta$.

These ratios can be written as:

$$BC^2 = AB \times BH \text{ and } AC^2 = AB \times AH. \tag{A.2}$$

Summing these two equalities, we obtain:

$$BC^2 + AC^2 = AB \times BH + AB \times AH = AB \times (AH + BH) = AB^2, \tag{A.3}$$

which, tidying up, is the Pythagorean theorem:

$$BC^2 + AC^2 = AB^2. \tag{A.4}$$

□

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Bibliography

Publications Related to the Thesis

All publications related to the thesis should be referenced in the text.

Other Publications (optional)

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Biography

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